

Yang Ni

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🎓 Education

Tsinghua University Ph.D. Candidate in Astronomy, Advisor: Prof. Xuening Bai	2022 – Present <i>Beijing, China</i>
Princeton University Visiting Student Research Collaborators (VSRC), Host Advisor: Prof. Jeremy Goodman	2026.02 – 2026.08 <i>New Jersey, US</i>
Nanjing University B.S. in Astronomy (Magna Cum Laude), Advisor: Prof. Bin Luo	2018 – 2022 <i>Nanjing, China</i>
Massachusetts Institute of Technology Summer Research (remote), Host Advisor: Prof. Mark Vogelsberger	2020 – 2021 <i>Cambridge, US</i>

🔬 Research Interests

- Early-stage evolution of protoplanetary disks; planet formation
- Star formation in AGN disks
- Evolution of giant molecular clouds; star formation in ISM

📄 Publications

(**F** for first author, **C** for co-author)

- F4.** *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks II. Accretion, Migration, Spin, and Internal Structure of GI Fragments,*
Y. Ni, H. Deng, X. Bai *To be submitted, arXiv:2609.37984*
- F3.** *Modeling the Evolution of Protoplanetary Disks: Two Pathways from Gravitational Instability to MHD Wind-Driven Accretion,*
Y. Ni, W. Xu, X. Bai *re-submitted to ApJ (minor revision), arXiv:2607.12007*
- F2.** *Radiation Hydrodynamics of Self-gravitating Protoplanetary Disks I. Direct Formation of Gas Giants via Disk Fragmentation,*
Y. Ni, H. Deng, X. Bai *ApJ, 995, 96*
- F1.** *The life cycle of giant molecular clouds in simulated Milky Way-mass galaxies,*
Y. Ni, H. Li, M. Vogelsberger, L. Sales, F. Marinacci, P. Torrey *A&A, 699, A282*
- C4.** *MHD-RIGEL: Impact of magnetic field on simulated dwarf galaxies,* J. Chu, H. Li, Y. Deng, & **Y. Ni.** *to be submitted*
- C3.** *RIGEL: Feedback regulated cloud-scale star formation efficiency in a simulated dwarf galaxy merger,* Y. Deng, ..., **Y. Ni**, et al. *A&A, 704, A240*
- C2.** *The THESAN-ZOOM project: Star formation efficiency from giant molecular clouds to galactic scale in high-redshift starbursts,*
Z. Wang, ..., **Y. Ni**, et al. *MNRAS, 544, 2675*
- C1.** *GRB 210121A: A Typical Fireball Burst Detected by Two Small Missions,*
X. I. Wang, ..., **Y. Ni**, et al. *ApJ, 922, 237*

Talks and Posters

Selected:

Protoplanetary Disks & Exoplanets 2026, The Kavli Institute for Astronomy and Astrophysics (KIAA) at Peking University, China	September 2026
(Invited talk) Purple Mountain Observatory, Jiangsu, China	August 2026
AthenaK Summer School, Penn State University, US	July 2026
Exoplanets and Planet Formation Mini-Workshop, Tsinghua University, China	December 2025
ICEPF Conference 2025, Shanghai, China (<i>Poster</i>)	December 2025
East Asia Numerical Astrophysics Meeting (EANAM10), Jeju, Korea	September 2025
Exoplanets & Planet Formation Workshop 2025, Yunnan, China	August 2025
NSFC Scientific Meeting on CPDs and Satellite Formation, Beijing, China	May 2024
Exoplanets & Planet Formation Workshop 2023, Beijing, China (<i>Poster</i>)	December 2023
International Conference on Resolving Galaxy Ecosystems Across All Scales, The Chinese University of Hong Kong, China	December 2023
Colloquium, Department of Astrophysics, Universität Zürich, Switzerland	November 2023

Technical Skills

- ✔ Contributor and experienced user of the multi-grid self-gravity module in AthenaK
- ✔ Contributor and experienced user of the M1-radiative transfer module in GIZMO (PLM reconstruction, RT subcycling, implicit source solver)
- ✔ Experienced user of the radiative (magneto-)hydrodynamic simulation with GIZMO
- ✔ Experienced User of IRAF/Pyraf; The Chandra Interactive Analysis of Observations (CIAO)
- ✔ Experienced User of Python: astropy, h5py, numpy, matplotlib, scipy, yt, networkx... ; C/C++; Mathematica; L^AT_EX
- ✔ Experienced User of AI agent harness including Claude Code and Codex
- ✔ Experienced User of LLMs including Claude Opus 4.6/4.7/4.8/5, Claude Fable 5, GPT 5.5/5.6/6, Gemini 3.1 Pro/3.5 Flash

Honors & Awards

Selected:

Tsinghua University Merit Student,	Tsinghua University
Tsinghua Scholarship for Overseas Graduate Studies,	Tsinghua University
Comprehensive Scholarship,	Tsinghua University
National Astronomical Observatory of China (NAOC) Scholarship,	NAOC
People's Scholarship First Prize,	The Ministry of Education of the PRC
Outstanding Undergraduate,	Nanjing University
"Top-notch" First Prize,	Nanjing University
G. Zheng Scholarship for Oversea Study,	Zheng Foundation

References

(PhD Advisor) **Prof. Xuening Bai**, Tsinghua University

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(PhD Co-Advisor) **Prof. Hongping Deng**, Shanghai Astronomical Observatory (SHAO)

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Prof. Jeremy Goodman, Princeton University

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Prof. Mark Vogelsberger, Massachusetts Institute of Technology

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Prof. Hui Li, Tsinghua University

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Prof. Bin Luo, Nanjing University

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